

INTELLIGENT DEAD RECKONING

SIH Problem Statement 26168

MASTER BLUEPRINT

Current-State Architecture + Frozen Decisions + Target-System Design

Purpose of this blueprint

This document is the implementation-level source of truth for the project as it exists on GitHub main at commit da84645. It integrates the verified Android acquisition/recording/export/replay/map foundation with the still-pending scientific navigation stack. It deliberately separates implemented evidence from target architecture so that future work cannot silently convert a prototype visualization into a navigation claim.

Repository: SaiChayank/intelligent-dead-reckoning

Baseline: main @ da84645 (Added Live Map)

Prepared: 25 September 2026

Document state: MASTER BLUEPRINT - CURRENT BASELINE

Document Control

Field	Value
Document	SIH PS26168 - Intelligent Dead Reckoning - Master Blueprint
Role	Architecture, integration, decision and acceptance baseline
Repository baseline	GitHub main @ da84645fc39f07363a58d3cee035bd196186f11d
Primary evidence	Repository source, tests and reports; current-state project documentation; frozen design decisions
Reference style	PS26145 decision-chain documents supplied by the project owner; structure reused, technical content not copied
Supersession rule	Repository evidence supersedes stale status statements in older planning material; official/frozen requirements are not silently altered
Next companion file	Planning & Implementation Plan (to be generated separately after this blueprint)

Authoritative-use rule

Use this blueprint for future implementation guidance. If later work conflicts with a frozen decision, the conflict must be raised explicitly with evidence and an impact assessment before changing the baseline.

Blueprint Map

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1. Source Hierarchy, Labels and Governance

This blueprint uses the same disciplined separation of requirement, evidence and team decision used in the supplied PS26145 document chain. It does not treat every historical statement as equally authoritative.

Priority	Source	Use
1	Official PS26168/SIH material already incorporated into project decisions	Problem objective and mandatory outcome constraints.
2	Frozen decisions approved during project design	Architecture, product, contract, lifecycle and evaluation choices.
3	Current GitHub repository and executed verification evidence	Truth for what is actually implemented and verified.
4	Supporting research / IO-VNBD evidence	Scientific constraints, dataset behavior and algorithm readiness.
5	Video/reference-inspired presentation ideas	Optional demo/UI inspiration only; never a requirement or benchmark result.

Label	Meaning
OFFICIAL / FROZEN TARGET	Problem-level target already accepted as binding for the project.
TEAM DECISION / FROZEN	Approved design decision to preserve unless explicitly revised.
VERIFIED	Executed test/device evidence exists for the stated scope.
IMPLEMENTED	Code exists, but verification may be narrower than production acceptance.
PARTIAL	Only part of the intended capability is delivered.
PLANNED	Approved future design; not current implementation.
BLOCKED	Work should not proceed past the gate until prerequisite evidence exists.
OPTIONAL / ENHANCEMENT	Useful addition outside the required core outcome.
ASSUMPTION	Unverified statement that must not be presented as fact.

Status discipline

The root README contains older status statements. Current-status claims in this blueprint follow the implemented repository modules and the newer RECORDING / EXPORT / REPLAY / OFFLINE_MAP / MAP_DEVICE_VERIFICATION evidence.

2. Problem Statement, Goals and Acceptance Definition

2.1 Core problem

SIH PS26168 targets vehicle navigation continuity when GNSS becomes unavailable, degraded or unreliable. The system must continue estimating vehicle motion and position from inertial information, limit accumulated drift, expose uncertainty, and recover cleanly when GNSS returns. The primary prototype must not depend on vehicle OBD/CAN data.

2.2 Frozen project targets

Target	Classification	Blueprint consequence
Positional drift during GNSS outage: target <10% of travelled distance	OFFICIAL / FROZEN TARGET	The final navigation stack is not acceptable until this is demonstrated on defined held-out outage scenarios.
Android application is the main prototype	TEAM DECISION / FROZEN	Live capture, inference, state display, recording and map integration converge in the Android product.
No mandatory cloud inference / runtime network dependency	FROZEN REQUIREMENT	Navigation must work on device; maps used for the prototype are locally packaged.

Target	Classification	Blueprint consequence
No OBD/CAN dependency for the primary solution	FROZEN REQUIREMENT	Phone/IMU + GNSS are sufficient inputs for the core prototype.
Approx. 10 Hz mobile navigation output	FROZEN DESIGN TARGET	Navigation output/presentation should be bounded near this cadence; sensor acquisition may be faster.
Higher-rate edge target around 200 Hz with FOG-class IMU	FROZEN DESIGN TARGET	Edge engine is secondary and reuses the navigation contracts/core after mobile correctness.

2.3 Final-system success definition

- Causal calibration and frame handling proven for the runtime sensor source; no dataset-specific shortcut masquerades as deployment calibration.
- Physically valid inertial mechanization with synthetic motion tests and stable initialization behavior.
- AI/learned correction improves held-out error relative to the classical baseline without leakage.
- Fusion responds to GNSS quality, continues through outages, carries calibrated confidence/uncertainty and rejoins GNSS without an unbounded visible jump.
- Runtime remains bounded in CPU, memory, queues and latency on the target phone.
- Offline map is a presentation consumer of navigation state; map matching and routing remain separate replaceable modules.
- Recorded/replayed sessions reproduce exact timestamps/source lineage for regression and evaluation.

3. Current Verified Baseline

Subsystem	Status	Current evidence / limitation
Phase 0 data audit	VERIFIED	Inventory, schema registry, synchronization evidence, raw integrity and synthetic tests exist.
Android real acquisition	VERIFIED	OnePlus 12R: accel/gyro/gravity/magnetometer, permissions, GNSS states, lifecycle, queues and 30-min endurance.
contracts/v1	VERIFIED	Strict Kotlin/Python codecs; exact Int64 timestamps; typed measurement/navigation payloads.
Recording contract	VERIFIED	Versioned metadata + measurements.jsonl contract in Kotlin/Python.
Foreground recorder	VERIFIED	10-minute monitored run: 179,358 records, zero recorder drops/write errors/ID gaps at final observation; recovery/failure tests.
Saved-session UI + local ZIP export	VERIFIED	SAF local export, exact-byte package, large-recording physical export checked.
Kotlin replay	VERIFIED	Read-only streaming replay with replay_real/replay_simulation, timing and lifecycle checks.
Offline MapLibre map	VERIFIED enhancement	Bundled Hyderabad MBTiles renders on device; no INTERNET permission; synthetic overlay presentation verified.
NavigationPresentation adapter	IMPLEMENTED	Consumes typed NavigationState and converts ENU+origin to WGS84 display state; not connected to real engine.
Production NavigationEngine	NOT IMPLEMENTED	Interface only.
Body-frame/calibration solution	BLOCKED/PARTIAL	Legacy IO-VNBD export semantics still prevent a universal defensible transform.
Classical INS	EXPERIMENTAL	Historical raw baseline shows ~382.7% drift/reference over 60 s; not acceptable.
AI correction	NOT IMPLEMENTED	No frozen training pipeline/model artifact.
EKF/UKF fusion	NOT IMPLEMENTED	No production estimator.

Subsystem	Status	Current evidence / limitation
Map matching / routing / guidance	NOT IMPLEMENTED	Rendering only; no road graph/routing engine.
Python Prompt-3 session layer	PARTIAL GAP	Contract codec exists; no dedicated exported-session reader/comparison package equivalent to Kotlin session layer.
Edge engine	NOT IMPLEMENTED	Target architecture only.

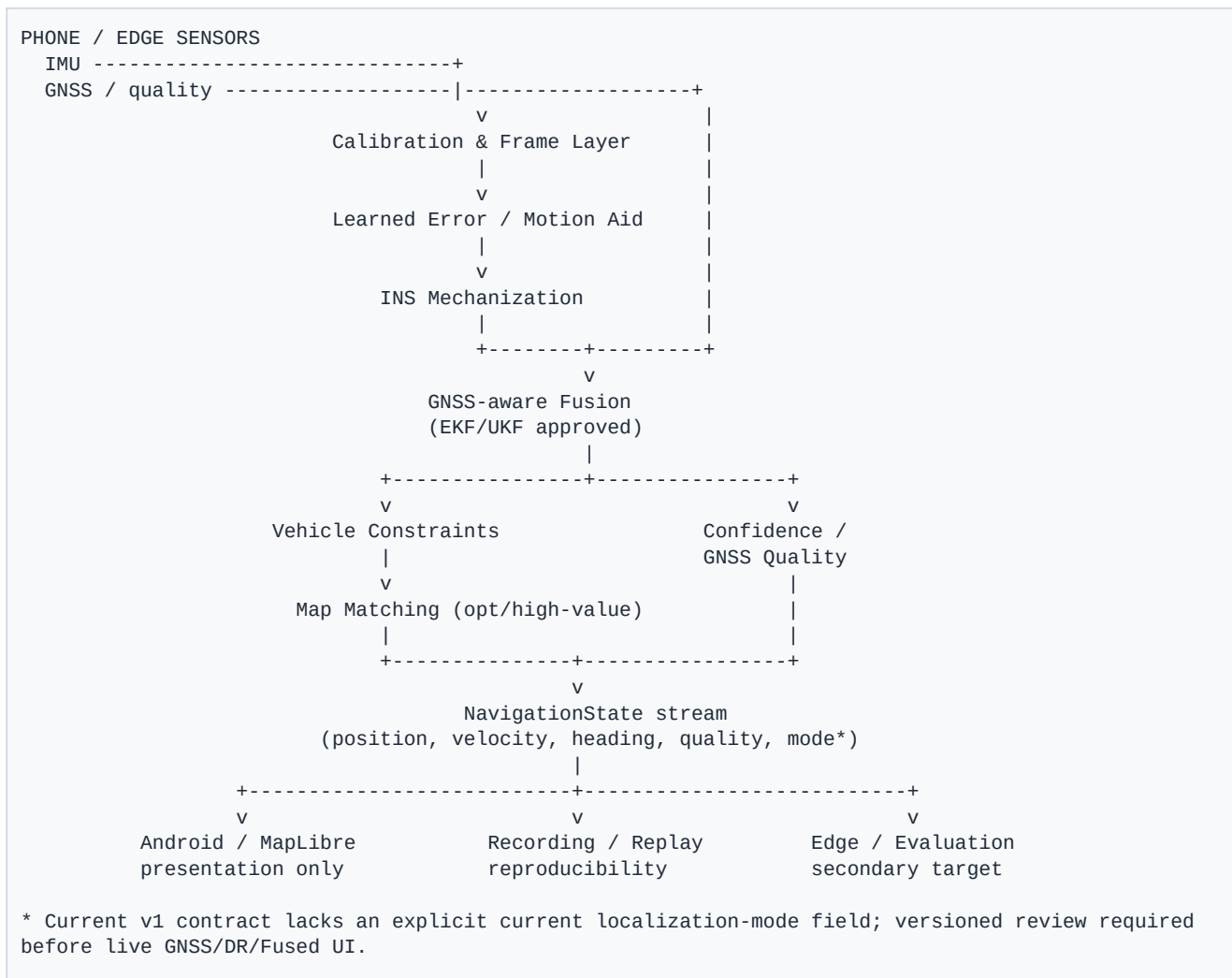
Baseline interpretation

The project has a strong acquisition/reproducibility/demo foundation but does not yet have a scientifically valid live dead-reckoning engine. The next algorithmic work begins at calibration/frame correctness, not at map animation.

4. Scope and Capability Classification

Capability	Class	Build decision
Real sensor/GNSS acquisition	CORE / VERIFIED	Keep frozen; regression-test only unless a defect is proven.
Recording, export, replay	CORE SUPPORT / VERIFIED	Keep as evaluation and reproducibility backbone.
Offline MapLibre rendering	OPTIONAL HIGH-VALUE / VERIFIED	Keep; connect to real NavigationState only after engine exists.
Synthetic outage/recovery map demo	OPTIONAL DEMO / VERIFIED	Keep clearly labelled synthetic; never use as performance evidence.
Calibration + frame resolution	CORE / BLOCKER	Must be resolved before production INS or AI training.
Classical INS	CORE	Rebuild/validate after frame/calibration gate.
AI error correction / learned motion aid	CORE	Train only after leakage-safe data pipeline and credible baseline.
Fusion / GNSS quality / recovery	CORE	Required for actual GNSS->DR->GNSS behavior.
Non-holonomic vehicle constraints	HIGH-VALUE	Add after INS/fusion state conventions are stable.
Map matching	HIGH-VALUE	Add after trustworthy navigation state; never use it to conceal poor inertial estimates.
Offline routing / rerouting	OPTIONAL	Prototype only if time remains; requires separate local road graph.
Turn-by-turn UI	OPTIONAL	Downstream of routing; not a localization algorithm.
Edge 200 Hz FOG target	FUTURE / SECONDARY	After shared core correctness and mobile demonstration.

5. Target End-to-End System Architecture



5.1 Non-negotiable architectural boundaries

- Map rendering does not estimate position. It consumes navigation output.
- Recording remains downstream of acquisition; disk I/O must never block sensor/location callbacks.
- Contracts remain UI-independent and cannot import Android/Compose/MapLibre classes.
- Replay never masquerades as live input and preserves source timestamps exactly.
- Offline training must not fit transforms or model parameters on held-out evaluation sequences.
- Map matching is a constraint/posterior stage, not a substitute for a valid INS/fusion estimator.
- Routing and turn-by-turn are downstream navigation services, separate from positioning/map matching.

6. Component Blueprints

6.1 Android acquisition

Input: Android accelerometer, gyroscope, gravity, magnetometer, location/GNSS status. Output: canonical Record stream. Preserve event and receipt monotonic timestamps, nullable GNSS values, sensor identity, provider, units, permission state, bounded queues and foreground lifecycle. Status: VERIFIED/FROZEN.

6.2 Measurement/navigation contract

contracts/v1 is the cross-language boundary. Source values real/simulation/replay_real/replay_simulation; exact decimal Int64 timestamps; strict nulls/enums/frames. NavigationState already supports origin+ENU position/velocity/heading/status/confidence companion events. Status: VERIFIED/FROZEN; explicit localization mode remains a known future schema decision.

6.3 Recording/export/replay

metadata.json + measurements.jsonl is canonical local-session storage. Recorder uses bounded async I/O, atomic metadata finalization and explicit recovery. Export ZIP is exact-byte packaging; replay is read-only and emits replay_* source. Status: VERIFIED/FROZEN within documented limits.

6.4 Calibration and frame resolution

Goal: causal phone/IMU-to-vehicle alignment, gyro bias and accelerometer treatment suitable for live mechanization. Legacy IO-VNBD exports cannot be assumed raw Android-device vectors. New live capture is the safer deployment calibration source. Status: BLOCKER before production INS.

6.5 INS mechanization

Implement a testable shared, UI-independent core with explicit frames, gravity handling, quaternion conventions, measured dt, gap handling and initialization modes. Must pass stationary/constant-velocity/turn/acceleration/bias/irregular-dt synthetic suites before real-data claims.

6.6 AI/learned correction

Purpose: reduce inertial/model error or estimate motion aid using causal features. Training must reuse the same feature definitions used at inference, freeze temporal/configuration holdouts, prevent future leakage, export a versioned on-device model, and demonstrate improvement over classical INS.

6.7 GNSS-aware fusion

Fuse inertial state with valid GNSS according to an explicit GNSS quality policy. During denial/state periods propagate DR; on recovery reintroduce measurement updates with gating/innovation handling so display transition is physically justified rather than cosmetically interpolated.

6.8 Vehicle constraints

Use non-holonomic constraints and motion sanity checks where valid: e.g., low lateral/vertical vehicle velocity assumptions. Must be gated by motion state and uncertainty; never force unrealistic constraints during slides, bumps or unusual motion.

6.9 Map matching

Use local road graph/geometry to score plausible road hypotheses from the fused navigation state and uncertainty. HMM/Viterbi or a simpler probabilistic snap can be evaluated. Keep raw/fused and matched positions separately traceable.

6.10 Map rendering

MapLibre renders bundled local OSM/OpenMapTiles-derived vector data. NavigationPresentation is the preferred adapter from NavigationState to display coordinates. Smooth marker interpolation is presentation-only and must not mutate engine timestamps/state.

6.11 Routing and guidance

Optional. Requires a routable local graph independent of MBTiles. Route planning, route progress, rerouting and turn instructions are downstream services; they may consume the matched position but never determine the estimator state.

6.12 Edge engine

Secondary target for ~200 Hz FOG-class IMU. Reuse core state conventions/contracts; platform-specific transport, sensor drivers and model runtime are replaceable. No edge implementation should fork the navigation mathematics from mobile.

7. Component Contract Matrix

Producer	Consumer	Contract / payload	Ownership / invariant
AndroidAcquisition	UI / LocalRecorder	contracts/v1 Record	Acquisition owns live sensor callbacks and monotonic event/receipt timestamps.
LocalRecorder	SessionFiles / Replay	recording/v1 metadata + measurements.jsonl	Recorder owns file creation/finalization/recovery; no semantic rewrite of measurements.

Producer	Consumer	Contract / payload	Ownership / invariant
SessionFiles	ExportController	Exact private file bytes	Export packages bytes; it does not reinterpret or validate navigation quality.
SessionFiles	ReplayReader	Read-only session stream	Replay validates canonical records and changes only emitted source to replay_*.
NavigationEngine (future)	NavigationPresentation	NavigationState + Confidence + GnssQualityState	Engine owns estimation; presentation owns conversion/display only.
Calibration layer	INS/Fusion	Versioned calibration state	Calibration ID/validity must be causally tied to the same mounting/session context.
AI model	INS/Fusion	Versioned model output + provenance	Training/inference feature parity; no hidden online learning in MVP.
Fusion	Map matcher	Pose, velocity, heading, covariance/confidence	Matcher constrains road hypothesis; retains raw fused state separately.
Map matcher	Routing/UI	Matched pose + road identity/progress	No feedback into estimator unless an explicitly designed measurement model is later approved.
Routing engine	Turn-by-turn UI	Route geometry + maneuver list	Routing has no sensor ownership and no localization authority.

Contract-gap decision

Before connecting the real engine to the UI status chip, explicitly decide whether v1 can represent GNSS/DR/Fused mode without ambiguity. If not, introduce a versioned contract revision rather than overloading NavigationStatus or gnss_used_after_initialization.

8. Data, Training and Model Blueprint

8.1 Data sources

Source	Purpose	Allowed use
IO-VNBD	Historical research, synchronized phone/VBOX comparisons, algorithm diagnostics	Use with documented timing/frame/unit limitations. Do not relabel world-like exports as raw Android device IMU.
Fresh Android recordings	Deployment-like sensor/GNSS sessions with canonical contracts	Preferred source for live sensor-rate/frame/lifecycle validation and future causal calibration experiments.
Synthetic motion fixtures	Mathematical correctness and estimator unit tests	Ground truth for stationary, straight, turns, accelerations, bias and outage logic.
VBOX/reference traces where available	Evaluation reference	Reference only; never allowed as deployable runtime input.

8.2 Data split and leakage policy

- Freeze train/validation/test splits before model tuning; split by route/session/driver/configuration, not random rows.
- Any alignment, normalization, calibration parameter or learned scaler must be fitted only on training/development data.
- Temporal windows may use only present/past information at inference time.
- No VBOX future value, future GNSS fix, or evaluation-sequence statistic may leak into deployable initialization or features.
- Report development/reference-assisted modes separately from deployable modes.

8.3 AI model acceptance

Gate	Required evidence
Baseline	Credible corrected classical INS on synthetic and real/reference data.
Target definition	Explicit output: e.g., speed/velocity correction, bias correction, or learned motion aid; not an ambiguous end-to-end black box.
Training integrity	Deterministic data manifest, split manifest, feature version, seed/config, model hash.
Generalization	Held-out route/configuration/driver evaluation where data supports it.
Improvement	Statistically and practically better drift/velocity error than the classical baseline on the same held-out windows.
Deployment	Exported ONNX/TFLite (chosen after runtime benchmark) with parity test between training runtime and Android inference.
Failure behavior	Out-of-distribution/unavailable model state does not fabricate high confidence; fallback path is explicit.

9. Runtime Pipeline and Navigation State Management

9.1 Live pipeline

1. Acquire asynchronous IMU/GNSS events and validate canonical contract semantics.
2. Apply causal calibration/frame transform to the relevant device-frame measurements.
3. Update learned correction/motion-aid state on bounded windows at its configured cadence.
4. Propagate INS at sensor cadence using measured monotonic dt; handle late/gap diagnostics explicitly.
5. Update GNSS quality state independently of whether a location payload exists.
6. When GNSS is valid, apply gated fusion measurement update; when stale/denied, continue DR propagation.
7. Apply vehicle constraints where their validity preconditions hold.
8. Emit navigation + confidence + GNSS-quality records at the mobile output cadence.
9. Optionally map-match the emitted fused state for road-aligned presentation/route progress.
10. Present/record/evaluate outputs without feeding UI smoothing back into estimator state.

9.2 GNSS -> DR -> GNSS recovery state behavior

Phase	Engine behavior	UI/map behavior
GNSS healthy	Fusion uses accepted GNSS measurements; inertial propagation continues between fixes.	Show fused/current position, quality and calibrated uncertainty.
GNSS degrading	Quality policy marks stale/degraded based on causal evidence; innovation/gating tightens as designed.	Status changes before hard outage where evidence supports it; avoid false "GPS lost" labels.
GNSS denied/unavailable	No fresh GNSS update. INS + AI aid + constraints propagate state and uncertainty.	Marker continues from engine output; mode labelled DR only if contract explicitly supports it.
GNSS returns	Validate fix freshness/quality; gate innovation; reintroduce measurements over estimator updates.	Display follows estimator convergence. Optional visual interpolation may smooth rendering but never changes engine state or hides large disagreement.
Recovery stable	Normal fused updates resume; outage metrics close and are recorded.	Return to normal fused/GNSS-supported status with continuity evidence.

9.3 Bounded-runtime rules

- No unbounded full-trip arrays in the live engine. Use bounded history required by algorithms only.
- Sensor callback path performs minimal mapping/validation; heavy work runs on owned workers/coroutines.
- Define explicit queue capacities, overflow policy and diagnostics for engine input/output.
- Do not propagate across unsafe IMU gaps without a defined reset/reinitialization policy.
- Use monotonic integer time as the integration clock; wall-clock time is metadata only.

10. Android Product Blueprint

Screen / area	Current / target responsibility
Dashboard	Source selection, acquisition controls, recording controls, high-level navigation/GNSS state and current position when engine exists.
Diagnostics	Sensor availability/rates, queue counters, permission/GNSS diagnostics, recorder/replay state, future engine timing/innovation/model diagnostics.
Saved Sessions	Session metadata, export, replay, completion/recovery status; already implemented.
Map	Offline MapLibre rendering, camera controls, synthetic demo today; later live engine presentation, travelled trail, confidence radius, matched position and route overlays.
Calibration	Future guided mount/calibration workflow with explicit validity and failure reasons.
Replay / Evaluation	Existing replay controls; later optional engine-through-replay analysis mode with deterministic output comparison.
About / Evidence	Scope labels, offline/privacy statement, version/build provenance and demo limitations.

UX rule

Every screen that can show synthetic/replay/live data must visibly distinguish the source. No synthetic drift visualization may be presented as measured model performance.

11. Offline Map, Map Matching, Routing and Guidance

11.1 Current map stack

Layer	Current choice	Status
Renderer	MapLibre Android OpenGL 13.6.1	VERIFIED
Base data	Bundled central-Hyderabad MBTiles derived from OpenStreetMap/OpenMapTiles/OpenFree Map snapshot	VERIFIED subset
Glyphs	Bundled local Noto Sans glyph ranges	VERIFIED subset
Network	MapLibre disconnected; INTERNET/NETWORK/WIFI permissions removed	VERIFIED on device
Presentation adapter	NavigationPresentation -> MapPresentation -> GeoJSON overlay	IMPLEMENTED
Live engine feed	NavigationEngine -> NavigationPresentation	NOT IMPLEMENTED

11.2 Separation of functions

Function	Definition	Current status
Map rendering	Draw roads/labels/buildings/overlays.	Implemented offline.
Positioning	Estimate pose from sensors/GNSS.	Not implemented beyond raw GNSS acquisition / synthetic demo.
Map matching	Select plausible road-aligned position from uncertain pose.	Not implemented.

Function	Definition	Current status
Routing	Find path between origin and destination on a routable road graph.	Not implemented.
Turn-by-turn	Track route progress and produce maneuver guidance/rerouting.	Not implemented.

11.3 Offline routing strategy

If routing is added for the polished prototype, use a separate locally packaged routable graph rather than treating MBTiles as routable data. Candidate engines should be evaluated for Android footprint, preprocessing complexity, offline licensing, graph size and rerouting latency. The blueprint does not freeze a routing engine yet because localization correctness remains higher priority.

- Package only the demo region to control APK/private-storage size.
- Prefer preprocessing on the development machine; runtime performs local graph search.
- Keep route geometry and maneuver metadata versioned independently of the map tile pack.
- Allow route progress to consume matched/fused position; never let routing override estimator coordinates.
- Attribution/licensing must remain visible and bundled notices retained.

12. Edge-Engine Blueprint

Concern	Decision
Role	Secondary deployment for high-rate FOG-class IMU / approximately 200 Hz target.
Language/runtime	Python is acceptable for research/control; production edge runtime should reuse exported model + shared math or a performant implementation with parity tests.
Contracts	Reuse the same conceptual event/state schema; platform-specific transport wrapper may differ.
Clock	Monotonic device clock with explicit boot/session identity; never merge clocks across devices without synchronization metadata.
Model	Same frozen model/version as mobile where inputs are compatible; otherwise separate hardware-domain model with explicit provenance.
Acceptance	Parity tests against reference implementation, throughput at target cadence, bounded latency/memory, no hidden cloud dependency.

13. Repository and Deployment Structure

13.1 Current repository

contracts/	strict Python/Kotlin measurement + recording contracts
docs/	historical project docs / proposals
mobile/	Android product, recording/export/replay/map
reports/	data/frame/INS/acquisition evidence
tests/	Python contract/foundation/data tests
training/	dataset audits, frame diagnostics, experimental INS
data/raw/	local only; ignored by Git
models/	future model artifacts; binaries ignored by Git
edge/	planned; not yet implemented
core/	planned shared navigation core; not yet implemented

13.2 Target additions

Target path	Responsibility
core/navigation/	Pure navigation math/state: frames, mechanization, fusion, constraints; no Android UI.

Target path	Responsibility
core/calibration/	Causal calibration models and validity state.
training/datasets/	Versioned loaders/splits/manifests separated from legacy exploratory scripts.
training/features/	Causal feature computation reused by training and export verification.
training/models/	Training/evaluation/export pipelines and model cards.
mobile/.../navigation/	Android adapter/worker that owns NavigationEngine lifecycle and feeds UI/recording.
mobile/.../matching/	Optional local map-matching module.
mobile/.../routing/	Optional offline route graph/engine adapter.
edge/	Secondary high-rate runtime after core stabilization.
reports/navigation/	Machine-readable benchmark outputs and acceptance evidence.

13.3 Build/deployment modes

Mode	Purpose
Developer workstation	Python audit/training/evaluation; Android Studio/Gradle; map-pack preprocessing.
Android debug demo	Primary SIH prototype; app-private recordings + bundled map; no mandatory network.
Replay verification	Deterministic session playback into readers/engine for regression.
Edge bench	Future high-rate IMU runtime and model/core parity testing.

14. Security, Privacy and Offline Controls

Control	Current / target behavior	Class
Network independence	Android manifest removes INTERNET/network/Wi-Fi permissions for current app; map resources are local.	VERIFIED
App-private storage	Recordings and installed map pack live under app-private no-backup storage.	VERIFIED
User-mediated export	SAF ACTION_CREATE_DOCUMENT, local provider filtering; no automatic upload.	VERIFIED
Path safety	Session IDs and pack paths validated; export ZIP has fixed entry names.	IMPLEMENTED
Data integrity	Map pack uses manifest SHA-256; recording codecs validate structure/counts.	VERIFIED/IMPLEMENTED
Model integrity	Future model artifact hash + version should be recorded in engine/session metadata.	PLANNED
Privacy	Diagnostics should not expose absolute private paths or raw coordinates unnecessarily.	FROZEN RULE
Background behavior	No background recording/service; background stops acquisition/recording/replay according to frozen lifecycle.	VERIFIED
Secrets	No API keys required for offline map/runtime; .env/keys ignored by Git.	VERIFIED design

15. Verification and Validation Strategy

Layer	Required tests
Contract	Kotlin/Python golden fixtures; invalid corpus; Int64 >2^53; nulls; versions; source/session invariants.
Acquisition	Sensor hardware mapping, rates, permissions, lifecycle, queue bounds, timestamps, missing hardware, GNSS provider/null semantics.
Recording/export/replay	Start/stop, overflow, write failures, recovery, exact ZIP bytes, replay source mapping, counts, corruption, lifecycle.
Frame/calibration	Rotation math, six-face/known-turn capture, stationary bias, remount invalidation, causal calibration windows.
INS	Stationary, constant velocity, constant acceleration, circular motion, turns, bias injection, irregular dt, gaps, coordinate conversions.
AI	Leakage-safe splits, train/serve feature parity, ablation vs baseline, calibration of confidence if claimed, model export parity.
Fusion	GNSS good/degraded/stale/denied transitions, innovation gates, outage propagation, reacquisition convergence, delayed/outlier fixes.
Map matching	Road ambiguity, intersections, parallel roads, U-turns, off-road, uncertainty growth and wrong-road recovery.
Map/routing UI	Offline render, overlays, attribution, no network fallback, route progress, lifecycle, performance.
System	End-to-end live and replay runs; bounded memory/queues; <10% drift evaluation; GNSS->DR->GNSS evidence.
Security	No network permission, local export only, path traversal rejection, model/data integrity checks.

16. Performance and Acceptance Criteria

16.1 Core acceptance metrics

Metric	Target / rule	Measurement method
GNSS-outage positional drift	<10% of travelled distance (frozen target)	Reference trajectory held-out outage segments; report final and RMS error plus drift/distance.
Mobile navigation output	Approx. 10 Hz target	Timestamp emitted NavigationState records; report p50/p95 cadence and missed updates.
Sensor ingestion	Maintain device-observed rates without unbounded queues	Per-channel measured rate + queue depth/high-water + drop/invalid counters.
Navigation latency	Bounded; explicit target to be frozen after first implementation benchmark	Input received_ns -> output received/emission time; p50/p95/p99.
Memory	No session-sized growth in live engine	Long foreground run with PSS/RSS + queue/history sizes.
Recovery continuity	No uncontrolled position jump	Record innovation/correction magnitude and display continuity; distinguish estimator correction from UI interpolation.
Offline behavior	No mandatory network access	Airplane/offline execution of core + map on target device.

16.2 MVP acceptance gate

- All previously verified acquisition/recording/export/replay/map regression suites still pass.
- Frame/calibration gate is passed for the runtime sensor source; no unsupported legacy-export transform is used.
- Classical INS passes synthetic mechanization tests and produces a physically credible real/reference baseline.
- AI model, if included in the MVP claim, improves the held-out baseline and passes train/serve parity.
- Fusion handles defined GNSS quality/outage/recovery scenarios causally and exposes uncertainty/confidence.

16. At least one real or controlled reference route demonstrates the frozen <10% outage-drift target under clearly defined conditions, or the prototype must explicitly state that the target is not yet met.
17. Offline Android demonstration runs without mandatory internet/cloud/OBD/CAN dependency.
18. Map display consumes engine output; synthetic and live modes remain visibly distinct.

17. SIH Demonstration Blueprint

Step	Demonstration	Evidence shown
1	Launch offline Android app	Build/version; no INTERNET permission; Hyderabad map loads locally.
2	Start real phone sensors	Real source banner; sensor names/rates; queue diagnostics.
3	Show GNSS healthy state	Provider/quality/accuracy fields when available; no fabricated speed/bearing.
4	Start local recording	Recording ID, counters, app-private storage, zero upload.
5	Begin navigation route	Engine state/position/heading/confidence once implemented; map consumes output.
6	Enter controlled GNSS outage	Quality transitions; GNSS updates stop; DR continues.
7	Show DR continuity	Marker/trail continues from engine; uncertainty grows; outage distance/time visible.
8	GNSS recovery	Validated fix reintroduced; estimator converges; correction/innovation visible.
9	Show drift result	Reference vs estimate and drift/distance metric for the outage.
10	Stop/finalize recording	Completed counts and no silent loss.
11	Export locally	SAF ZIP; exact metadata + JSONL.
12	Replay session	replay_real source; deterministic counts/timestamps; no hardware listener activation.
13	Optional synthetic/reference demo	Video-inspired comparison overlays only if clearly labelled synthetic.

Demo fallback policy

If the real navigation algorithm is not yet scientifically accepted, demonstrate the verified acquisition/recording/replay/offline-map foundation and the synthetic map scenario separately. Do not convert the synthetic outage path into a fake “working AI DR” claim.

18. Risk Register and Mitigations

Risk	Severity	Mitigation / gate
Legacy IO-VNBD frame semantics cannot support a universal raw-device transform	BLOCKER	Use fresh canonical Android captures for deployable calibration; keep legacy data for bounded research/reference.
Classical INS currently catastrophically wrong	HIGH	Rebuild after frame gate; synthetic mechanization suite before AI.
AI trained before credible baseline / leakage controls	HIGH	Block model training acceptance until splits/features/baseline are frozen.
GNSS/DR/Fused mode absent from v1 contract	MEDIUM	Versioned schema review before live map status integration.
Map matching hides estimator error	HIGH	Always retain/display/evaluate raw fused pose separately from matched pose.
Routing expands scope and storage	MEDIUM	Keep optional; local demo-region graph only after navigation core passes.

Risk	Severity	Mitigation / gate
APK/map-pack size grows	MEDIUM	Regional packs, app-private install, optional pack modularization later.
Battery/thermal load from continuous high-rate processing	MEDIUM	Profile sustained runs; decouple sensor cadence from UI cadence; optimize model/inference after correctness.
Replay/session cap insufficient for long trips	LOW/MEDIUM	Current 1M-record limit is explicit; future disk-backed index or session segmentation if needed.
Single-device verification overfits assumptions	MEDIUM	Add second Android device/API level before production claim.
Stale documentation misstates implementation status	MEDIUM	Update root README/status files at each frozen baseline; blueprint supersedes older current-state wording.

19. Frozen Decision Register

ID	Decision	State
D01	Android/Kotlin/Jetpack Compose is the primary prototype product.	FROZEN
D02	Python is the offline research/training/evaluation environment.	FROZEN
D03	Raw IO-VNBD data stays outside Git and must not be modified.	FROZEN
D04	contracts/v1 is the current canonical typed measurement/navigation exchange contract.	FROZEN
D05	Recording uses metadata.json + measurements.jsonl; source timestamps remain exact.	FROZEN
D06	Acquisition and recording are foreground-only; no silent restart after background.	FROZEN
D07	Recorder I/O is downstream; sensor/location callbacks never block on disk.	FROZEN
D08	Replay emits replay_real/replay_simulation and never masquerades as live.	FROZEN
D09	Offline MapLibre is a navigation-output consumer, not a localization engine.	FROZEN
D10	Current offline map stack uses local OSM/OpenMapTiles/OpenFreeMap-derived data and preserves attribution.	FROZEN CURRENT CHOICE
D11	Synthetic map demo remains explicitly synthetic until a real NavigationEngine exists.	FROZEN
D12	No silent unit/frame conversion may be introduced to make INS appear successful.	FROZEN
D13	AI/EKF/map matching cannot be used cosmetically to hide unresolved mechanization correctness.	FROZEN
D14	No OBD/CAN dependency for the primary solution.	FROZEN
D15	Runtime has no mandatory cloud inference/network dependency.	FROZEN
D16	Final target is <10% positional drift during defined GNSS outages.	FROZEN TARGET
D17	Mobile navigation output target is approximately 10 Hz; edge target approximately 200 Hz.	FROZEN DESIGN TARGET

ID	Decision	State
D18	Future routing/turn-by-turn remains optional and separate from localization/map matching.	FROZEN SCOPE RULE

20. Build Gates and Blueprint Readiness

Gate	Entry condition	Exit evidence	Current
G0 Foundation	Repository/contracts/data audit	Strict contracts + reproducible evidence	PASS
G1 Android acquisition	Foundation pass	Physical sensor/GNSS/lifecycle/endurance evidence	PASS
G2 Recording reproducibility	Acquisition pass	Recording/recovery/export/replay device evidence	PASS
G3 Offline map presentation	Android foundation	Offline renderer/device evidence; synthetic labels	PASS (enhancement)
G4 Frame/calibration	Canonical live/raw-device evidence	Approved vehicle-frame transform/bias/calibration procedure	OPEN / BLOCKER
G5 Classical INS	G4 pass	Synthetic mechanization suite + credible real/reference baseline	BLOCKED BY G4
G6 AI correction	G5 pass + split/feature plan	Held-out improvement + export parity	BLOCKED BY G5
G7 Fusion/recovery	G5/G6 inputs defined	GNSS quality/outage/recovery tests + confidence	BLOCKED
G8 Vehicle constraints/map matching	Fusion state stable	Constraint + road-ambiguity evaluation	BLOCKED
G9 Live map integration	Real NavigationState stream	Verified GNSS/DR/recovery presentation	BLOCKED
G10 Final SIH acceptance	Core chain complete	<10% outage-drift evidence + end-to-end demo	BLOCKED
G11 Edge target	Shared core stable	~200 Hz edge throughput/parity evidence	FUTURE

MASTER BLUEPRINT READINESS

READY AS THE IMPLEMENTATION SOURCE OF TRUTH. The architecture, verified foundation, current gaps and future component boundaries are sufficiently defined to plan execution. This does NOT mean the final navigation solution is ready; algorithmic execution is blocked at the frame/calibration gate.

20.1 Immediate next planning focus

19. Preserve and regression-test the verified Android acquisition/recording/export/replay/map baseline.
20. Create the dedicated Planning & Implementation Plan from this blueprint, beginning with the calibration/frame-resolution gate.
21. Add a reusable Python Prompt-3 session reader/comparison layer so fresh phone sessions can be analyzed with the same canonical semantics.
22. Build the shared navigation core with synthetic mechanization tests before returning to AI training.
23. Only after the classical baseline is credible, define/train the AI correction model and fusion stages.
24. Connect the real NavigationState stream to the existing NavigationPresentation/MapLibre path last, without moving localization logic into the UI.

Appendix A - Evidence and Repository Sources

Artifact	Role in blueprint
mobile/ACQUISITION.md	Verified Android acquisition behavior and physical acceptance.

Artifact	Role in blueprint
contracts/v1/README.md + Kotlin/Python codecs	Canonical measurement/navigation contract semantics.
contracts/recording/v1/*	Canonical recording metadata/session contract.
mobile/RECORDING.md	Recorder architecture, sustained run, recovery/storage evidence.
mobile/EXPORT.md	Saved-session/export architecture and physical verification.
mobile/REPLAY.md	Replay semantics and physical verification.
mobile/OFFLINE_MAP.md	Offline map package, renderer and scope.
mobile/MAP_DEMO_FEATURES.md	Reference-video feature provenance and synthetic demo boundaries.
mobile/MAP_DEVICE_VERIFICATION.md	Physical map acceptance on OnePlus 12R.
reports/frame_resolution_followup.md	Current frame/calibration scientific limitations.
reports/phase1_metrics.md	Historical experimental raw INS failure metrics.
reports/foundation_readiness.md	Earlier gate audit; historical implementation statements superseded where repository has since advanced.
SIH_PS26168_Project_Documentation_Current_State.docx	Companion current-state documentation used for consolidated requirements/status.
Supplied PS26145 reference ZIP	Structure/discipline reference for decision chain, classification, audit and verification organization only.

Appendix B - Claims Boundary

Safe current claim	Unsafe / unsupported claim
The app physically acquires phone IMU and permitted location/GNSS data on the tested OnePlus 12R.	The app currently performs production dead reckoning during GNSS loss.
The app records, exports and replays canonical local sessions.	Fresh phone recording validates the legacy IO-VNBD exporter.
The app renders a bundled offline Hyderabad map and synthetic GNSS/DR/recovery presentation.	The red comparison trace is measured INS/AI drift.
A NavigationPresentation adapter can display valid NavigationState records.	A production NavigationEngine currently emits those states.
Historical experimental INS drift is very large and frame correctness remains unresolved.	The historical INS is a credible baseline ready for AI optimization.
Map matching/routing are planned separable components.	The current road overlay performs road snapping or routing.

END OF MASTER BLUEPRINT

Next companion document: Planning & Implementation Plan